

Index

Unit – 1 $\rightarrow$ Descriptive Statistics	3
1) Method 2 $\rightarrow$ Measure of Central Tendency	3
2) Method 3 $\rightarrow$ Measure of Variability	8
3) Method 4 $\rightarrow$ Moments	10
4) Method 5 $\rightarrow$ Measures of Skewness	12
5) Method 6 $\rightarrow$ Kurtosis	13

Unit – 1 $\rightsquigarrow$ Descriptive Statistics

Method 2 $\rightsquigarrow$ Measure of Central Tendency

Example of Method-2.1: Examples of Mean

A	1	Find mean of following data: (a) 2, 8, 4, 6, 10, 12, 4, 8, 14, 16 (b) 10, 9, 21, 16, 14, 18, 20, 18, 14, 18, 23, 16, 18, 4 Answer: (a) 8.4, (b) 15.6429																												
A	2	Find the mean for following data: <table><tr><td>Weight of students</td><td>18</td><td>22</td><td>30</td><td>35</td><td>39</td><td>42</td><td>45</td><td>47</td></tr><tr><td>Number of students</td><td>4</td><td>5</td><td>8</td><td>8</td><td>16</td><td>4</td><td>2</td><td>3</td></tr></table> Answer: 34.5	Weight of students	18	22	30	35	39	42	45	47	Number of students	4	5	8	8	16	4	2	3										
Weight of students	18	22	30	35	39	42	45	47																						
Number of students	4	5	8	8	16	4	2	3																						
A	3	Find the mean for following data: <table><tr><td>x</td><td>10</td><td>20</td><td>36</td><td>40</td><td>50</td><td>56</td><td>60</td><td>70</td><td>72</td><td>80</td><td>88</td><td>92</td><td>95</td></tr><tr><td>f</td><td>1</td><td>1</td><td>3</td><td>4</td><td>3</td><td>2</td><td>4</td><td>4</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td></tr></table> Answer: 59.3	x	10	20	36	40	50	56	60	70	72	80	88	92	95	f	1	1	3	4	3	2	4	4	1	1	2	3	1
x	10	20	36	40	50	56	60	70	72	80	88	92	95																	
f	1	1	3	4	3	2	4	4	1	1	2	3	1																	
B	4	Find the mean if survey regarding the weights (kg) of 45 students of class X of a school was conducted and the following data was obtained: <table><tr><td>x</td><td>20 – 25</td><td>25 – 30</td><td>30 – 35</td><td>35 – 40</td><td>40 – 45</td><td>45 – 50</td><td>50 – 55</td></tr><tr><td>f</td><td>2</td><td>5</td><td>8</td><td>10</td><td>7</td><td>10</td><td>3</td></tr></table> Answer: 38.83	x	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55	f	2	5	8	10	7	10	3												
x	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55																							
f	2	5	8	10	7	10	3																							

B	5	The following data represents the no. of foreign visitors in a multinational company in every 10 days during last 2 months. Use the data to find the mean. <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>No. of visitors</td><td>12</td><td>18</td><td>27</td><td>20</td><td>17</td><td>06</td></tr></table> Answer: 28	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	No. of visitors	12	18	27	20	17	06																
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60																										
No. of visitors	12	18	27	20	17	06																										
C	6	Find the missing frequency from the following data if mean is 19.92. <table><tr><td>Class</td><td>4 – 8</td><td>8 – 12</td><td>12 – 16</td><td>16 – 20</td><td>20 – 24</td></tr><tr><td>f</td><td>11</td><td>13</td><td>16</td><td>14</td><td>?</td></tr><tr><td></td><td>24 – 28</td><td>28 – 32</td><td>32 – 36</td><td>36 – 40</td><td></td></tr><tr><td></td><td>9</td><td>17</td><td>6</td><td>4</td><td></td></tr></table> Answer: 9	Class	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24	f	11	13	16	14	?		24 – 28	28 – 32	32 – 36	36 – 40			9	17	6	4							
Class	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24																											
f	11	13	16	14	?																											
	24 – 28	28 – 32	32 – 36	36 – 40																												
	9	17	6	4																												
C	7	Find the mean of the following frequency distribution: <table><tr><td>Mid value</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td><td>40</td><td>45</td><td>50</td><td>55</td></tr><tr><td>Frequency</td><td>2</td><td>22</td><td>19</td><td>14</td><td>3</td><td>4</td><td>6</td><td>1</td><td>1</td></tr><tr><td>Cumulative</td><td>2</td><td>24</td><td>43</td><td>57</td><td>60</td><td>64</td><td>70</td><td>71</td><td>72</td></tr></table> Answer: 27.8472	Mid value	15	20	25	30	35	40	45	50	55	Frequency	2	22	19	14	3	4	6	1	1	Cumulative	2	24	43	57	60	64	70	71	72
Mid value	15	20	25	30	35	40	45	50	55																							
Frequency	2	22	19	14	3	4	6	1	1																							
Cumulative	2	24	43	57	60	64	70	71	72																							

Example of Method-2.2: Median

A	1	Find the median of following data: (a) 6, 20, 43, 50, 19, 53, 0, 37, 78, 1, 15. (b) 10, 34, 27, 24, 12, 27, 20, 18, 15, 30. (c) 110, 115, 108, 112, 120, 116, 140, 135, 128, 132. Answer: (a) 20, (b) 22, (c) 118																		
A	2	If the median of the data is 2, find the value of a: −9, −4, a, 5, 8, 11. Answer: − 1																		
A	3	Obtain the median size of shoes sold from the following data: <table border="1"><tr><td>Size</td><td>5</td><td>5.5</td><td>6</td><td>6.5</td><td>7</td><td>7.5</td><td>8</td></tr><tr><td>Pair</td><td>30</td><td>40</td><td>50</td><td>150</td><td>300</td><td>600</td><td>950</td></tr></table> Answer: 8.32	Size	5	5.5	6	6.5	7	7.5	8	Pair	30	40	50	150	300	600	950		
Size	5	5.5	6	6.5	7	7.5	8													
Pair	30	40	50	150	300	600	950													
C	4	Calculate the missing frequency from the following distribution, it is being given that the median of the distribution is 24. <table border="1"><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>No. of students</td><td>5</td><td>25</td><td>x</td><td>18</td><td>7</td></tr></table> Answer: x = 25	Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	No. of students	5	25	x	18	7						
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50															
No. of students	5	25	x	18	7															
C	5	The following table gives the marks obtained by 50 students in mathematics. Find the median. <table border="1"><tr><td>x</td><td>10–14</td><td>15–19</td><td>20–24</td><td>25–29</td><td>30–34</td><td>35–39</td><td>40–44</td><td>45–49</td></tr><tr><td>f</td><td>4</td><td>6</td><td>10</td><td>5</td><td>7</td><td>3</td><td>9</td><td>6</td></tr></table> Answer: 29.5	x	10–14	15–19	20–24	25–29	30–34	35–39	40–44	45–49	f	4	6	10	5	7	3	9	6
x	10–14	15–19	20–24	25–29	30–34	35–39	40–44	45–49												
f	4	6	10	5	7	3	9	6												

Example of Method-2.3: Mode

A	1	Find the mode of following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>f</td><td>8</td><td>10</td><td>11</td><td>16</td><td>20</td><td>25</td><td>15</td><td>9</td><td>6</td></tr></table> Answer: 6	x	1	2	3	4	5	6	7	8	9	f	8	10	11	16	20	25	15	9	6
x	1	2	3	4	5	6	7	8	9													
f	8	10	11	16	20	25	15	9	6													
A	2	Find the mode from the following frequency distribution: <table><tr><td>x</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>f</td><td>5</td><td>6</td><td>8</td><td>7</td><td>9</td><td>8</td><td>9</td><td>6</td></tr></table> Answer: 12 & 14	x	8	9	10	11	12	13	14	15	f	5	6	8	7	9	8	9	6		
x	8	9	10	11	12	13	14	15														
f	5	6	8	7	9	8	9	6														
B	3	Find the mode of following data: <table><tr><td>Class</td><td>200 – 220</td><td>220 – 240</td><td>240 – 260</td><td>260 – 280</td></tr><tr><td>f</td><td>7</td><td>15</td><td>21</td><td>19</td></tr><tr><td></td><td>280 – 300</td><td>300 – 320</td><td>320 – 340</td><td></td></tr><tr><td></td><td>6</td><td>4</td><td>2</td><td></td></tr></table> Answer: 255	Class	200 – 220	220 – 240	240 – 260	260 – 280	f	7	15	21	19		280 – 300	300 – 320	320 – 340			6	4	2	
Class	200 – 220	220 – 240	240 – 260	260 – 280																		
f	7	15	21	19																		
	280 – 300	300 – 320	320 – 340																			
	6	4	2																			
B	4	Find the mode of following data: <table><tr><td>Class</td><td>400 – 500</td><td>500 – 600</td><td>600 – 700</td><td>700 – 800</td><td>800 – 900</td></tr><tr><td>f</td><td>8</td><td>16</td><td>20</td><td>17</td><td>3</td></tr></table> Answer: 657.14	Class	400 – 500	500 – 600	600 – 700	700 – 800	800 – 900	f	8	16	20	17	3								
Class	400 – 500	500 – 600	600 – 700	700 – 800	800 – 900																	
f	8	16	20	17	3																	
C	5	The mode of the following data is 67. Find the missing frequency x. <table><tr><td>Amount</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td><td>80 – 90</td></tr><tr><td>Frequency</td><td>5</td><td>x</td><td>15</td><td>12</td><td>7</td></tr></table> Answer: 8	Amount	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	Frequency	5	x	15	12	7								
Amount	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90																	
Frequency	5	x	15	12	7																	
B	6	An insurance company obtained the following data for accident claims (in thousand rupees) from a particular region. Find its mean, median and mode. <table><tr><td>Amount</td><td>1 – 3</td><td>3 – 5</td><td>5 – 7</td><td>7 – 9</td><td>9 – 11</td><td>11 – 13</td></tr><tr><td>Frequency</td><td>6</td><td>47</td><td>75</td><td>46</td><td>18</td><td>8</td></tr></table> Answer: $\bar{x}$ = 6.47, M = 6.2533, Z = 6.96	Amount	1 – 3	3 – 5	5 – 7	7 – 9	9 – 11	11 – 13	Frequency	6	47	75	46	18	8						
Amount	1 – 3	3 – 5	5 – 7	7 – 9	9 – 11	11 – 13																
Frequency	6	47	75	46	18	8																

C	7	Obtain the mean, median and mode for the following information:				
		Marks	0 <	10 <	20 <	30 <
		Number of Students	50	38	20	5
		Answer: $\bar{x} = 17.6$, $M = 17.2222$, $Z = 16.6667$				

Method 3 $\rightsquigarrow$ Measure of Variability

Example of Method-3: Dispersion

A	1	Find the standard deviation for the following distribution:																				
<table><tr><td>x</td><td>5</td><td>15</td><td>25</td><td>30</td></tr><tr><td>f</td><td>2</td><td>1</td><td>1</td><td>3</td></tr></table>			x	5	15	25	30	f	2	1	1	3										
x	5	15	25	30																		
f	2	1	1	3																		
Answer: 10.6904																						
C	2	Find the standard deviation for the following distribution:																				
<table><tr><td>Class</td><td>0 – 100</td><td>100 – 200</td><td>200 – 300</td><td>300 – 400</td></tr><tr><td>f</td><td>6</td><td>10</td><td>18</td><td>20</td></tr><tr><td></td><td>400 – 500</td><td>500 – 600</td><td>600 – 700</td><td>700 – 800</td></tr><tr><td></td><td>15</td><td>12</td><td>10</td><td>9</td></tr></table>			Class	0 – 100	100 – 200	200 – 300	300 – 400	f	6	10	18	20		400 – 500	500 – 600	600 – 700	700 – 800		15	12	10	9
Class	0 – 100	100 – 200	200 – 300	300 – 400																		
f	6	10	18	20																		
	400 – 500	500 – 600	600 – 700	700 – 800																		
	15	12	10	9																		
Answer: 196.21																						
C	3	Find the standard deviation and variance of the mark distribution of 30 students at mathematics examination in a class as below:																				
<table><tr><td>Class</td><td>10 – 25</td><td>25 – 40</td><td>40 – 55</td><td>55 – 70</td><td>70 – 85</td><td>85 – 100</td></tr><tr><td>f</td><td>2</td><td>3</td><td>0</td><td>14</td><td>8</td><td>3</td></tr></table>			Class	10 – 25	25 – 40	40 – 55	55 – 70	70 – 85	85 – 100	f	2	3	0	14	8	3						
Class	10 – 25	25 – 40	40 – 55	55 – 70	70 – 85	85 – 100																
f	2	3	0	14	8	3																
Answer: $\sigma = 19.3391$, $V = 374.0008$																						
B	4	Runs scored by two batsmen A, B in 9 consecutive matches is given below:																				
<table><tr><td>A</td><td>85</td><td>20</td><td>62</td><td>28</td><td>74</td><td>5</td><td>69</td><td>4</td><td>13</td></tr><tr><td>B</td><td>72</td><td>4</td><td>15</td><td>30</td><td>59</td><td>15</td><td>49</td><td>27</td><td>26</td></tr></table>			A	85	20	62	28	74	5	69	4	13	B	72	4	15	30	59	15	49	27	26
A	85	20	62	28	74	5	69	4	13													
B	72	4	15	30	59	15	49	27	26													
Which of the batsman is more consistent?																						
Answer: Batsman B is more consistent.																						
C	5	Goals scored by two team A and B in a football season were as shown in the table. Find out which team is more consistent.																				
<table><tr><td>Number of goals in a match</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Team A</td><td>27</td><td>9</td><td>8</td><td>5</td><td>4</td></tr><tr><td>Team B</td><td>17</td><td>9</td><td>6</td><td>5</td><td>3</td></tr></table>			Number of goals in a match	0	1	2	3	4	Team A	27	9	8	5	4	Team B	17	9	6	5	3		
Number of goals in a match	0	1	2	3	4																	
Team A	27	9	8	5	4																	
Team B	17	9	6	5	3																	
Answer: Team B is more consistent.																						

C	6	Lives of two models of refrigerators turned in for new models in a recent survey are given in the adjoining table.																					
		<table><tr><td>Life (in year)</td><td>0 – 2</td><td>2 – 4</td><td>4 – 6</td><td>6 – 8</td><td>8 – 10</td><td>10 – 12</td></tr><tr><td>Model A</td><td>5</td><td>16</td><td>13</td><td>7</td><td>5</td><td>4</td></tr><tr><td>Model B</td><td>2</td><td>7</td><td>12</td><td>19</td><td>9</td><td>1</td></tr></table>	Life (in year)	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12	Model A	5	16	13	7	5	4	Model B	2	7	12	19	9	1
		Life (in year)	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12															
		Model A	5	16	13	7	5	4															
Model B	2	7	12	19	9	1																	
(1) What is the average life of each model of these refrigerators?																							
(2) Which model shows more uniformity?																							
		Answer: (1) 5.12 & 6.16, (2) Model B																					
B	7	Find the mean deviation about the mean, median and mode for the following data: 5, 10, 17, 20, 23, 20.																					
		Answer: MD($\bar{x}$) = 5.5556, MD(M) = 5.1667, MD(Z) = 5.1667																					
B	8	Find mean deviation about the mean, median and mode for the following data: <table><tr><td>x</td><td>2</td><td>5</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>f</td><td>2</td><td>8</td><td>10</td><td>7</td><td>8</td><td>5</td></tr></table>	x	2	5	6	8	10	12	f	2	8	10	7	8	5							
x	2	5	6	8	10	12																	
f	2	8	10	7	8	5																	
		Answer: MD($\bar{x}$) = MD(M) = MD(Z) = 2.3																					
C	9	Find mean deviation about the mean, median and mode for the following data: <table><tr><td>Class</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>f</td><td>5</td><td>10</td><td>20</td><td>9</td><td>6</td></tr></table>	Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	f	5	10	20	9	6									
Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60																		
f	5	10	20	9	6																		
		Answer: MD($\bar{x}$) = 8.28, MD(M) = 8.20, MD(Z) = 8.30																					

Method 4 $\rightsquigarrow$ Moments

Example of Method-4: Moments

A	1	Find the first four moments about assumed mean 14, actual mean and zero for the data 11, 12, 14, 16, 20. Answer: $\mu = 0,$ 10.24, 19.162, 213.5872 $\mu' = 0.6,$ 10.6, 37.8, 281.8 $v = 14.6,$ 223.4, 3579.8, 57225.8														
B	2	Calculate the four moments about assumed mean 15, actual mean and zero for following distribution. <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>f</td><td>6</td><td>10</td><td>14</td><td>6</td><td>4</td></tr></table> Answer: $\mu = 0,$ 34, 40.5, 2707 $\mu' = -1,$ 35, -62.5, 2750 $v = 14,$ 230, 4212.5, 83375	x	5	10	15	20	25	f	6	10	14	6	4		
x	5	10	15	20	25											
f	6	10	14	6	4											
B	3	Calculate the moments about assumed mean 3, actual mean and zero for following distribution: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f</td><td>5</td><td>4</td><td>3</td><td>7</td><td>1</td><td>1</td></tr></table> Answer: $\mu = 0,$ 2.0862, 0.5017, 9.0299 $\mu' = -0.0952,$ 2.0952, -0.0952, 8.9524 $v = 2.9048,$ 10.5238, 43.1905, 191.6667	x	1	2	3	4	5	6	f	5	4	3	7	1	1
x	1	2	3	4	5	6										
f	5	4	3	7	1	1										

C	4	Calculate the moments about assumed mean 35, actual mean and zero for following distribution:																
		<table><tr><td>x</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td></tr><tr><td>f</td><td>8</td><td>12</td><td>20</td><td>30</td><td>15</td><td>10</td><td>5</td></tr></table>	x	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	f	8	12	20	30	15	10	5
x	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70											
f	8	12	20	30	15	10	5											
		Answer: $\mu = 0$, 236.76, 264.336, 141290.0876 $\mu' = -1.8$, 240, -1020, 144000 $v = 33.2$, 1339, 60440, 2957125																
C	5	Calculate the moments about assumed mean 65, actual mean and zero for following distribution:																
		<table><tr><td>Class</td><td>60 – 62</td><td>63 – 65</td><td>66 – 68</td><td>69 – 71</td><td>72 – 74</td></tr><tr><td>f</td><td>5</td><td>18</td><td>42</td><td>27</td><td>8</td></tr></table>	Class	60 – 62	63 – 65	66 – 68	69 – 71	72 – 74	f	5	18	42	27	8				
Class	60 – 62	63 – 65	66 – 68	69 – 71	72 – 74													
f	5	18	42	27	8													
		Answer: $\mu = 0$, 8.5275, -2.6933, 199.3759 $\mu' = 2.45$, 14.53, 74.69, 516.13 $v = 67.45$, 4558.03, 308586.79, 20930221.03																

Method 5 $\rightsquigarrow$ Measures of Skewness

Example of Method-5: Skewness

B	1	Find skewness by the method of moments for data: 38.2, 40.9, 39.5, 44, 39.6, 40.5, 39.5. Answer1. 1418																
B	2	Karl Pearson’s coefficient of skewness of a distribution is 0.3, its variance is 8 and mean is 200. Find the mode and median for the distribution. Answer: 199.15, 199.72																
B	3	From the marks scored by 120 students in section A and 120 students in section B of a class, the following measures were obtained <table border="1"><tr><td>Section A</td><td>$\mu_A = 46.83$</td><td>$\sigma_A = 14.8$</td><td>Mode = 51.67</td></tr><tr><td>Section A</td><td>$\mu_B = 47.83$</td><td>$\sigma_B = 14.8$</td><td>Mode = 47.07</td></tr></table> Determine which distribution of marks is more skewed. Answer: Section A is more skewed	Section A	$\mu_A = 46.83$	$\sigma_A = 14.8$	Mode = 51.67	Section A	$\mu_B = 47.83$	$\sigma_B = 14.8$	Mode = 47.07								
Section A	$\mu_A = 46.83$	$\sigma_A = 14.8$	Mode = 51.67															
Section A	$\mu_B = 47.83$	$\sigma_B = 14.8$	Mode = 47.07															
B	4	Find skewness of the following data using method of moment. <table border="1"><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr><tr><td>f</td><td>8</td><td>15</td><td>20</td><td>32</td><td>23</td><td>17</td><td>5</td></tr></table> Answer: – 0.1067	x	5	10	15	20	25	30	35	f	8	15	20	32	23	17	5
x	5	10	15	20	25	30	35											
f	8	15	20	32	23	17	5											
C	5	Find Karl Pearson’s coefficient of skewness and skewness based on the method of moments for the following data: <table border="1"><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>f</td><td>13</td><td>20</td><td>30</td><td>25</td><td>12</td></tr></table> Answer: – 0.0925, 0.82	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	f	13	20	30	25	12				
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50													
f	13	20	30	25	12													

Method 6 $\rightsquigarrow$ Kurtosis

Example of Method-6: Kurtosis

A	1	Find the Kurtosis for the data 1, 3, 7, 9, 10. Also comment on type of distribution. Answer: Kurtosis = 1.45, Distribution is platykurtic																		
C	2	Find the kurtosis for the following data. Also, Comment on type of distribution. <table border="1"><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>f</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table> Answer: Kurtosis = 2.2583, Distribution is platykurtic	Class	0 – 10	10 – 20	20 – 30	30 – 40	f	1	3	4	2								
Class	0 – 10	10 – 20	20 – 30	30 – 40																
f	1	3	4	2																
C	3	Find out the kurtosis of the following data: <table border="1"><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>f</td><td>1</td><td>4</td><td>3</td><td>2</td></tr></table> Answer: 2.102	Class	0 – 10	10 – 20	20 – 30	30 – 40	f	1	4	3	2								
Class	0 – 10	10 – 20	20 – 30	30 – 40																
f	1	4	3	2																
C	4	Find the coefficient of variation, β_1 and β_2 for the following data: <table border="1"><tr><td>x</td><td>170 – 180</td><td>180 – 190</td><td>190 – 200</td><td>200 – 210</td><td>210 – 220</td><td>220 – 230</td><td>230 – 240</td><td>240 – 250</td></tr><tr><td>f</td><td>52</td><td>68</td><td>85</td><td>92</td><td>100</td><td>95</td><td>70</td><td>28</td></tr></table> Answer: C. V. = 9.4, $\beta_1 = 0.0034$, $\beta_2 = 2.0340$	x	170 – 180	180 – 190	190 – 200	200 – 210	210 – 220	220 – 230	230 – 240	240 – 250	f	52	68	85	92	100	95	70	28
x	170 – 180	180 – 190	190 – 200	200 – 210	210 – 220	220 – 230	230 – 240	240 – 250												
f	52	68	85	92	100	95	70	28												

***** End of the Unit *****